

Simulation Based Inference (SBI)

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Goal

Infer the posterior distribution of unknown parameters θ after observing data x_o , based on a model of the data-generating process.

The data-generating process may be described by

- a likelihood $p(x \mid \theta)$, or
- a simulator from which we can generate $x \sim p(x \mid \theta)$.

Bayes' Theorem:

$$p(\theta \mid x_o) = \frac{p(x_o \mid \theta) p(\theta)}{p(x_o)} \propto p(x_o \mid \theta) p(\theta)$$

Goal

Generate samples from the posterior

$$p(\theta \mid x_o) \propto p(x_o \mid \theta)p(\theta).$$

Metropolis–Hastings algorithm:

- 1 Choose an initial parameter value $\theta^{(0)}$.
- 2 For $t = 1, \dots, T$:

- 1 Propose

$$\theta' \sim q(\theta' \mid \theta^{(t-1)}).$$

- 2 Compute the acceptance probability

$$\alpha = \min \left\{ 1, \frac{p(x_o \mid \theta')p(\theta')q(\theta^{(t-1)} \mid \theta')}{p(x_o \mid \theta^{(t-1)})p(\theta^{(t-1)})q(\theta' \mid \theta^{(t-1)})} \right\}.$$

- 3 Accept θ' with probability α ; otherwise retain $\theta^{(t-1)}$.

Repeated likelihood evaluations

- The likelihood must be available and sufficiently cheap to evaluate.

New Inference Run for Each Observation

Each new observation x_o generally requires a new MCMC run.

Sometimes, the data-generating process is represented by a simulator.

$$\theta \xrightarrow{\text{simulator}} x, \quad x \sim p(x \mid \theta)$$

- We can generate observations x for given parameters θ .
- However, the likelihood $p(x \mid \theta)$ may not be evaluable.

Simulation is possible; likelihood evaluation is not.

ABC approximates the posterior by retaining parameters that generate data similar to the observation x_o .

$$\theta_i \sim p(\theta), \quad x_i \sim p(x \mid \theta_i)$$

$$\text{Accept } \theta_i \iff d(s(x_i), s(x_o)) \leq \varepsilon$$

The accepted parameters approximate

$$p_\varepsilon(\theta \mid d(s(x), s(x_o)) \leq \varepsilon).$$

$s(\cdot)$: summary statistic, $d(\cdot, \cdot)$: distance, ε : tolerance

Comparison must be designed

- Requires informative summary statistics $s(x)$.
- Requires a meaningful, properly scaled distance d .

High-dimensional data

Acceptance rates collapse as the dimension of x or $s(x)$ grows:

$\varepsilon \downarrow \Rightarrow$ greater accuracy, lower acceptance rate.

New Inference Run for Each Observation

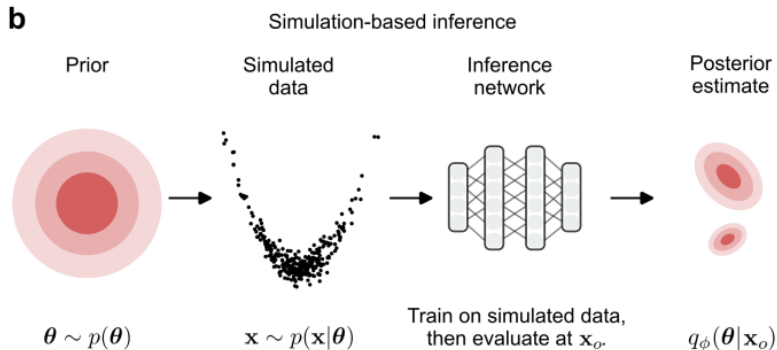
Each new observation x_o generally requires a new ABC run.

Instead of matching simulations to each observation separately, neural SBI learns the relationship between observations and parameters.

Training: $\{(\theta_i, x_i)\}_{i=1}^N \longrightarrow q_\phi(\theta \mid x)$

Inference: $x_o \longrightarrow q_\phi(\theta \mid x_o) \approx p(\theta \mid x_o)$

One training procedure enables inference for many observations.



Neural simulation-based inference workflow.

Source: Figure 1b in Deistler et al. (2025).

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A simulator generally consists of:

- ① **Prior** — distribution over the parameters θ to be inferred.
- ② **Process model** — description of the underlying data-generating process.
- ③ **Observation model** — description of how the process is observed or measured.

$$\theta \xrightarrow{\text{process model}} z \xrightarrow{\text{observation model}} x$$

The prior defines the plausible parameter values used for simulation:

$$\theta \sim p(\theta)$$

- Encodes prior knowledge about plausible parameters.
- Defines the parameter region over which inference is learned.
- Too narrow: plausible posterior regions may be excluded.
- Too broad: simulations are wasted on irrelevant regions.

The process model encodes the system and its dynamics:

$$z \sim p(z \mid \theta)$$

For example, a system with two interacting states:

$$z_{1,t+1} = z_{1,t} + \theta_1 z_{1,t} - \theta_2 z_{1,t} z_{2,t},$$

$$z_{2,t+1} = z_{2,t} - \theta_3 z_{2,t} + \theta_4 z_{1,t} z_{2,t}.$$

The observation model describes how the latent process z is measured:

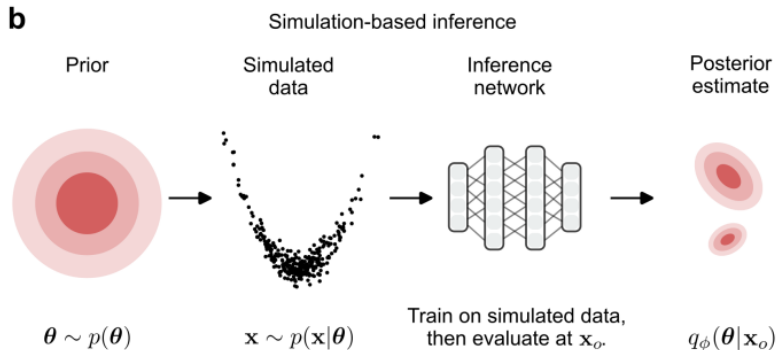
$$x \sim p(x | z)$$

For example, add measurement noise:

$$x_t = z_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \Sigma)$$

Correlated noise can model co-movement between measurements:

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$



Neural simulation-based inference workflow.

Source: Figure 1b in Deistler et al. (2025).

A normalizing flow transforms base noise through conditional, invertible layers:

$$u_0 \sim p_0(u), \quad u_k = f_{\phi,k}(u_{k-1}; h(x)), \quad \theta = u_K$$

- u_0 : sample from a simple base distribution
- p_0 : base distribution, e.g. $\mathcal{N}(0, I)$
- $h(x)$: representation of the observation
- $f_{\phi,k}$: k -th conditional, invertible transformation
- ϕ : learned network parameters
- $u_K = \theta$: resulting posterior sample

Input: $u_0 \sim p_0(u), \quad x \text{ or } h(x)$

Output: $\theta = f_\phi(u_0; x \text{ or } h(x)), \quad q_\phi(\theta | x)$

Loss:
$$\min_{\phi} \left\{ -\mathbb{E}_{p(\theta, x)} [\log q_\phi(\theta | x)] \right\}$$
$$\iff \min_{\phi} \mathbb{E}_{p(x)} [D_{\text{KL}}(p(\theta | x) \| q_\phi(\theta | x))]$$

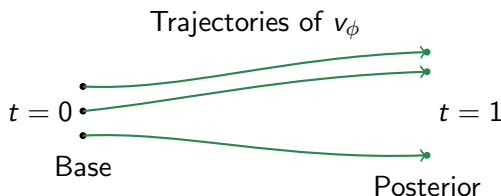
All layers are trained jointly: only the final distribution $q_\phi(\theta | x)$ is matched to the simulator posterior.

FMPE learns an observation-conditioned, time-dependent vector field:

$$\frac{d\theta_t}{dt} = v_\phi(\theta_t, t, x), \quad \theta_0 = u_0 \sim p_0(u)$$

Integrating from $t = 0$ to $t = 1$ transforms base noise into a posterior sample:

$$u_0 \xrightarrow[t:0 \rightarrow 1]{\text{integrate } v_\phi} \theta_1 \sim q_\phi(\theta \mid x)$$



For training, sample

$$\theta_1 \sim p(\theta), \quad x \sim p(x \mid \theta_1), \quad t \sim p(t), \quad \theta_t \sim p_t(\theta_t \mid \theta_1).$$

Define the path and its known target velocity:

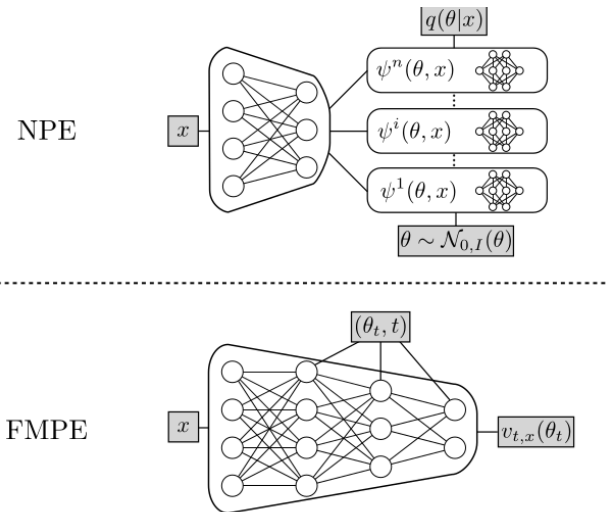
$$\theta_t = (1 - t)u_0 + t\theta_1, \quad \dot{\theta}_t = \theta_1 - u_0.$$

Input: $\theta_t, \quad t, \quad x$ or $h(x)$

Output: $v_\phi(\theta_t, t, x)$

$$\textbf{Loss: } \mathcal{L}_{\text{FMPE}}(\phi) = \mathbb{E} \left[\|v_\phi(\theta_t, t, x) - u_t(\theta_t \mid \theta_1)\|_2^2 \right]$$

Match local velocities $\implies$ recover the posterior flow



Comparison of NPE and FMPE network architectures.

Source: Figure 1 in Dax et al. (2023).

	NPE	FMPE
Architecture	Invertible transformations with tractable Jacobians.	Unrestricted vector-field network.
Network size	Scaling must preserve the specialized flow structure.	Width, depth and conditioning can be scaled more freely.
Inference time	Typically faster: direct transformation through the flow.	Typically slower: repeated network evaluations during ODE integration.

The population NPE loss can be written as

$$\mathcal{L}_{\text{NPE}}(\phi) = \text{const.} + \mathbb{E}_{p(x)} [D_{\text{KL}}(p(\theta | x) \| q_{\phi}(\theta | x))].$$

Therefore, with

- an increasing number of simulations $N \rightarrow \infty$,
- a sufficiently expressive conditional density model,
- consistent minimization of the population loss,

we obtain

$$\mathbb{E}_{p(x)} \left[D_{\text{KL}} \left(p(\theta | x) \| q_{\hat{\phi}_N}(\theta | x) \right) \right] \longrightarrow 0.$$

NPE directly minimizes divergence from the simulator posterior.

FMPE minimizes the discrepancy between the target and learned vector fields:

$$\text{MSE}_p(u, v_\phi) = \mathbb{E} \left[\|u_t - v_\phi\|_2^2 \right].$$

Under suitable regularity conditions,

$$D_{\text{KL}}(p_1 \| q_{\phi,1}) \leq C \text{MSE}_p(u, v_\phi)^{1/2}.$$

Therefore, with sufficient simulations and model capacity,

$$\text{MSE}_p(u, v_{\hat{\phi}_N}) \rightarrow 0 \quad \implies \quad D_{\text{KL}}(p_1 \| q_{\hat{\phi}_N,1}) \rightarrow 0.$$

FMPE approaches the posterior by recovering its transport field.

SBI replaces direct posterior computation with a learned approximation:

$$q_{\phi}(\theta \mid x_o) \approx p(\theta \mid x_o).$$

However,

low training or validation loss $\nRightarrow$ correct posterior at x_o

Possible reasons include:

- finite simulations and limited network capacity,
- errors averaged over x that hide failures near x_o ,
- an observation outside the simulated training distribution,
- prior or simulator misspecification.

The true posterior is generally unavailable for comparison. We therefore need complementary diagnostics for calibration, informativeness, structure and predictive behavior.

Based on the validation workflow in Deistler et al. (2025).

1. Point-estimate accuracy

Is the posterior mean or median close to the generating parameter?

2. Posterior calibration

Do credible probabilities have their advertised frequencies?

3. Informativeness

Does conditioning on x reduce uncertainty relative to the prior?

4. Joint structure

Are dependencies, correlations and multiple modes represented?

5. Predictive adequacy

Do posterior simulations resemble the observed data?

6. Reference-posterior agreement

Does the learned posterior match a trusted reference posterior?

No single diagnostic establishes posterior correctness.

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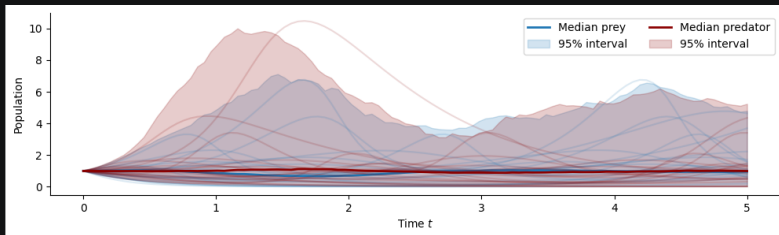
Ecological model

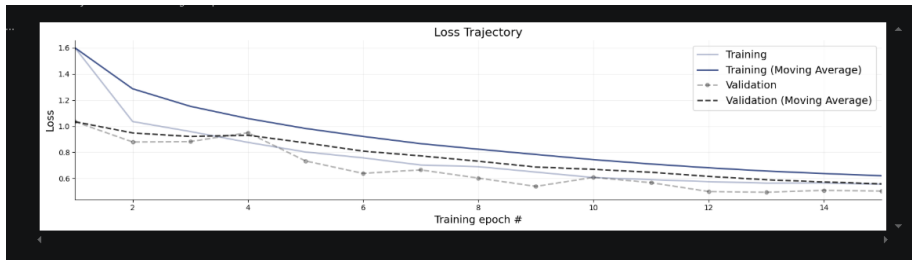
Let $x(t)$ be the prey population and $y(t)$ the predator population. Their dynamics are

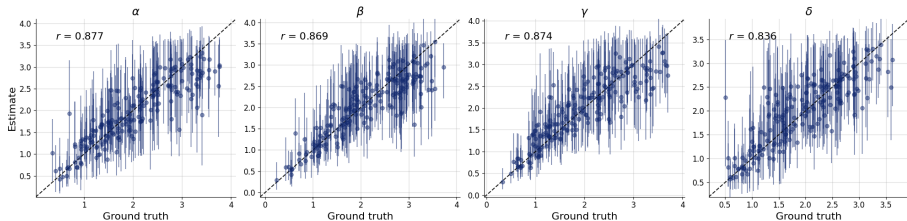
$$\frac{dx}{dt} = \alpha x - \beta xy, \quad \frac{dy}{dt} = -\gamma y + \delta xy.$$

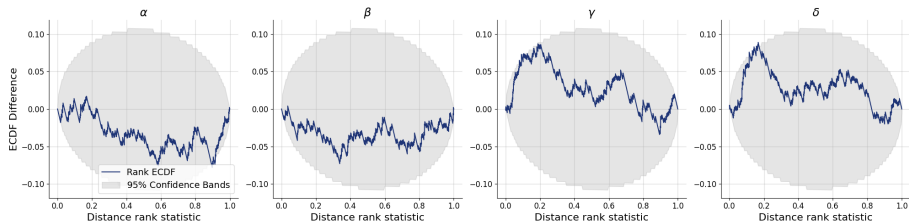
Symbol	Definition
$x(t)$	prey population at time t
$y(t)$	predator population at time t
α	prey growth rate in the absence of predators
β	predation rate: loss of prey through encounters
γ	predator death rate in the absence of prey
δ	predator growth rate from prey encounters

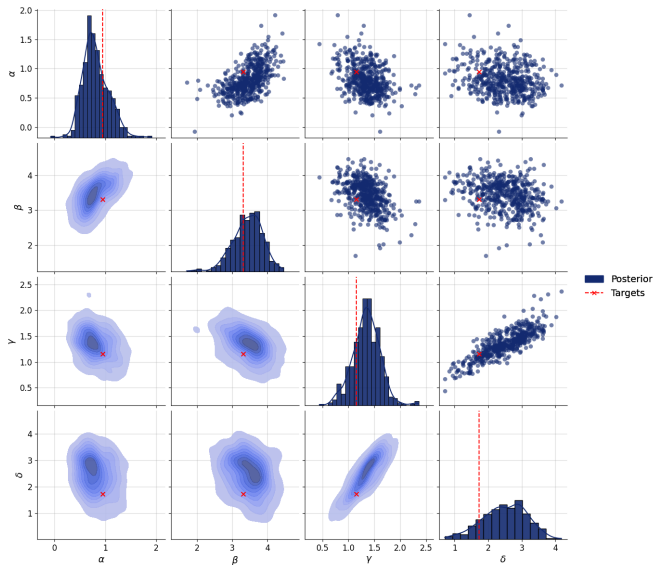
We infer $\theta = (\alpha, \beta, \gamma, \delta)$. The simulation starts at $(x_0, y_0) = (1, 1)$ and runs from $t = 0$ to $t = 5$. Each parameter has a scaled logit-normal prior on $(0.1, 4.0)$. The observation model retains every tenth time point and adds independent Gaussian noise with standard deviation 0.1.

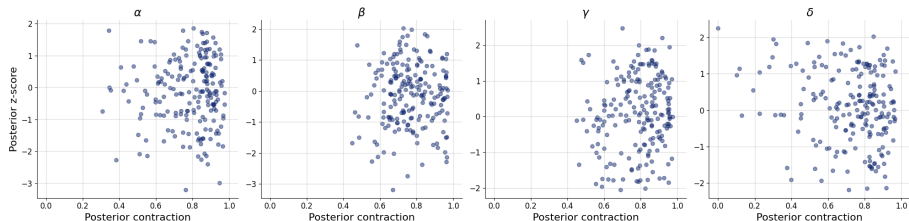


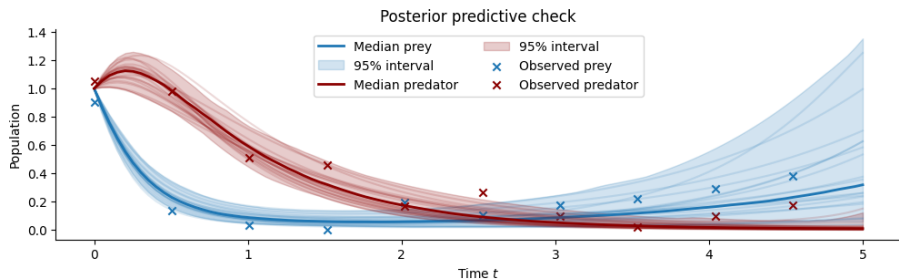




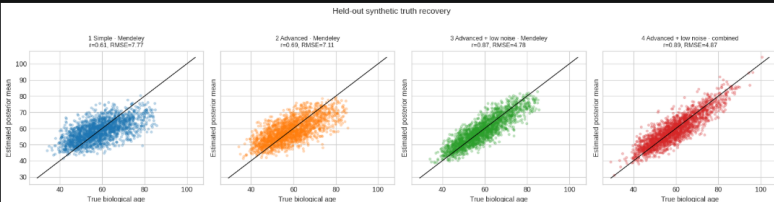


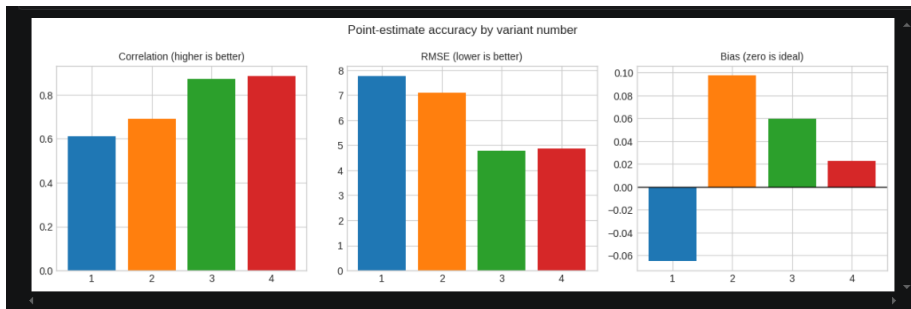


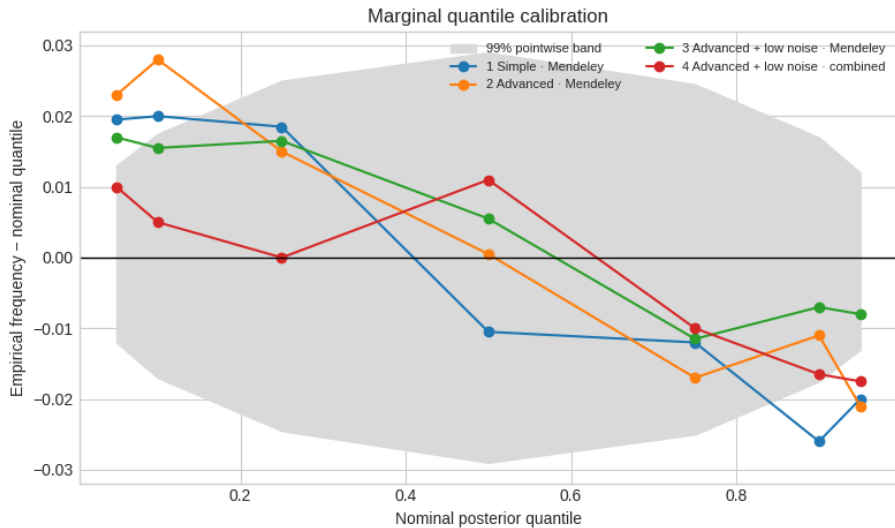




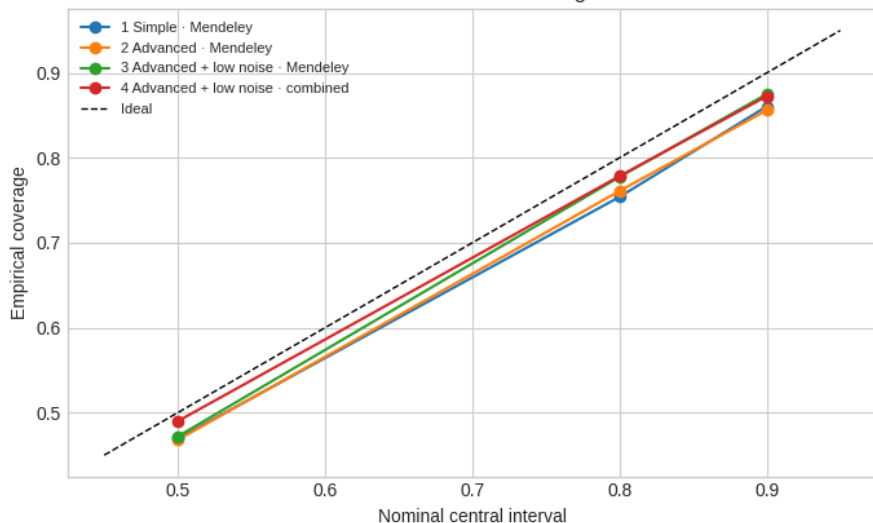
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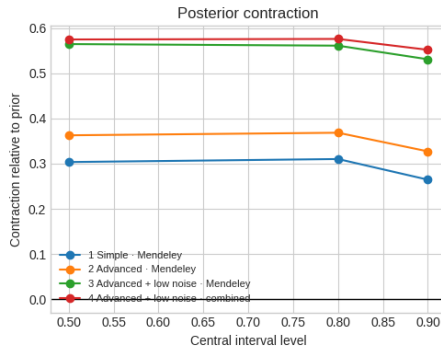
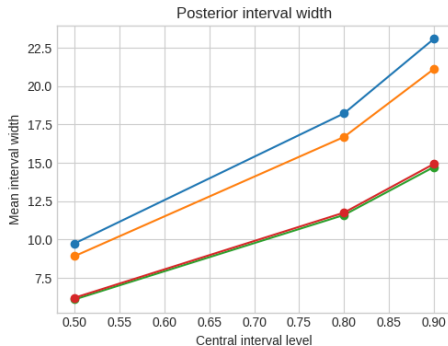




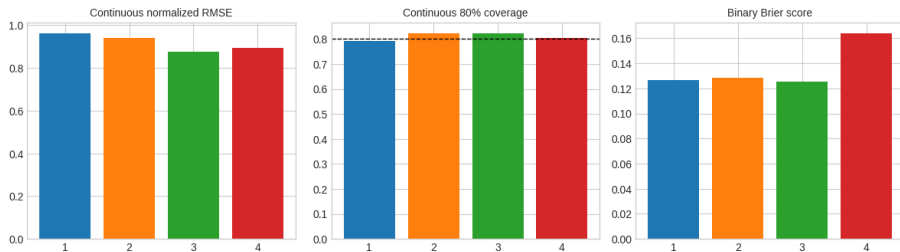


Credible-interval coverage





Aggregate posterior-predictive diagnostics by variant number



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	Reusable component	For each new observation
MCMC	Model and likelihood	Run a new Markov chain
ABC	Simulator	Generate simulations and repeat matching
NPE/FMPE	Simulator and trained conditional inference network	Evaluate the trained network; no new simulations or training

$$\textbf{NPE:} \quad (u_0, x_o) \xrightarrow{\text{forward pass}} \theta \sim q_\phi(\theta \mid x_o)$$

$$\textbf{FMPE:} \quad (\theta_t, t, x_o) \xrightarrow{\text{forward pass}} v_\phi(\theta_t, t, x_o) \xrightarrow{\text{ODE}} \theta \sim q_\phi(\theta \mid x_o)$$

The trained inference network is reused across observations.

Upfront cost

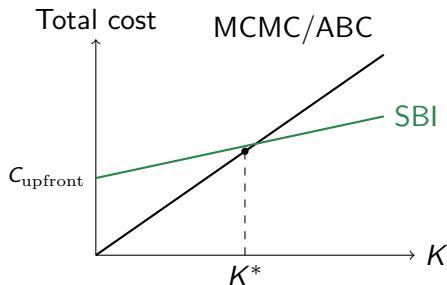
Classical inference

Amortized SBI

$$C_{\text{upfront}} = C_{\text{sim}} + C_{\text{train}} + C_{\text{validate}}$$

$$C_{\text{classical}}(K) = K C_{\text{per observation}}$$

$$C_{\text{SBI}}(K) = C_{\text{upfront}} + K C_{\text{infer}}$$



Break-even point:

$$K^* = \frac{C_{\text{upfront}}}{C_{\text{per observation}} - C_{\text{infer}}}$$

Amortization pays off for $K > K^*$, provided $C_{\text{per observation}} > C_{\text{infer}}$.

Neural SBI can first be viewed as an improvement in likelihood-free inference:

- learned representations $h(x)$ instead of handcrafted summaries;
- no manually chosen distance or acceptance tolerance;
- no rejection step with collapsing acceptance rates;
- direct learning of a posterior distribution or transport field.

Amortization is most valuable when:

① **Inference is repeated**

Many observations require posteriors under the same model.

② **Per-observation inference is expensive**

Classical inference requires many likelihood evaluations or simulations.

③ **The modeling world remains stable**

Simulator, prior and observation process can be shared.

④ **Future observations are covered**

New observations resemble those represented during training.

Examples: sensor streams, patient cohorts and repeated scientific events.

- Binary-black-hole parameters generate a waveform that is measured as a high-dimensional, noisy signal by multiple detectors.
- Although the likelihood is available, classical inference requires a new MCMC or nested-sampling run for every event and may take hours to months.
- One inference network can be trained on simulated waveforms and reused across events, reducing inference time to less than a minute (Deistler et al., 2025; Dax et al., 2023).

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The inference network must represent a complete joint distribution:

$$q_{\phi}(\theta \mid x), \quad \theta \in \mathbb{R}^d.$$

As d increases, it becomes harder to learn:

- dependencies between parameters;
- symmetries, trade-offs and multiple modes;
- sufficient coverage from a finite simulation budget.

Statistical convergence is relative to the simulator:

$$q_{\phi}(\theta \mid x) \longrightarrow p_{\text{sim}}(\theta \mid x).$$

However, for real observations,

$$p_{\text{sim}}(\theta \mid x) \neq p_{\text{real}}(\theta \mid x)$$

may hold.

- More simulations reduce approximation error.
- They cannot correct an incorrect simulator, observation model or prior.
- The wrong simulator posterior may be approximated increasingly faithfully.

SBI can converge accurately to the wrong target.

See Deistler et al. (2025).

The true posterior is generally unavailable, and high-dimensional joint posteriors cannot be visualized directly.

- Validation therefore often relies on one- and two-dimensional marginals.
- Correct marginals can hide incorrect dependencies, trade-offs or missing modes.
- Joint diagnostics become increasingly difficult and expensive as dimensionality grows.

See Deistler et al. (2025).

A trained inference network learns the inverse problem for one fixed modeling world M :

$$\mathcal{I}_{\phi}^{(M)} : x \mapsto q_{\phi}^{(M)}(\theta \mid x).$$

The modeling world fixes:

- the simulator and prior;
- the parameterization;
- the observation and noise model;
- the input representation.

Changing M generally requires new simulations, training and validation.

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Current SBI learns a separate inference rule for each simulator M :

$$\mathcal{I}_{\phi_M}^{(M)} : x_o \mapsto q_{\phi_M}(\theta_M \mid x_o).$$

The broader vision is one network conditioned on a simulator context:

$$C_M = \{(\theta_i, x_i)\}_{i=1}^m, \quad \mathcal{I}_{\phi}(C_M, x_o) \mapsto q_{\phi}(\theta_M \mid x_o, C_M).$$

- Begin with families of closely related simulators.
- Reuse recurring physical mechanisms and dependence structures.
- Specify a new inference task through examples in the context.
- Perform inference for a related task without updating the network weights.

Hypothesis: shared structure allows capacity to grow more slowly than a collection of separate inference networks.

1 Define a simulator family

Start with related simulators that share mechanisms and observation formats.

2 Encode variable context

Use a permutation-invariant set or attention architecture for a variable number of context pairs (θ_i, x_i) .

3 Represent variable parameter spaces

Represent parameters as tokens or use a schema and mask so that the network returns the correct number of posterior dimensions.

4 Test the scaling hypothesis

Compare network size, simulation cost and posterior quality as more simulators are added, including generalization to held-out tasks.

Does one shared network scale better than $\sum_M$ specialist networks?

Thanks for your attention!

Maximilian Dax, Jonas Wildberger, Simon Buchholz, Stephen R. Green, Jakob H. Macke, and Bernhard Schölkopf. 2023. Flow matching for scalable simulation-based inference. In *Advances in Neural Information Processing Systems*, volume 36.

Michael Deistler, Jan Boelts, Peter Steinbach, Guy Moss, Thomas Moreau, Manuel Glocckler, Pedro L. C. Rodrigues, Julia Linhart, Janne K. Lappalainen, Benjamin Kurt Miller, Pedro J. Gonçalves, Jan-Matthis Lueckmann, Cornelius Schröder, and Jakob H. Macke. 2025. Simulation-based inference: A practical guide. *arXiv preprint arXiv:2508.12939*.